B1 Flows, Fluctuations and Complexity, 2019 — Model Solutions

Question 1 — Vorticity, circulation and Couette flow

Problem. Define vorticity and circulation; treat solid-body rotation and the line vortex; derive the tangential Couette flow between two rotating cylinders, its vorticity and pressure; discuss the Reynolds number and evaluate numerically.

(a) Vorticity and circulation

Vorticity:

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}.$$

Physical meaning: $\boldsymbol{\omega}$ is twice the local angular velocity of a fluid element; it measures local spin.

Circulation around a closed contour C bounding surface S:

$$\Gamma = \oint_C \mathbf{u} \cdot \mathrm{d}\boldsymbol{\ell} = \int_S \boldsymbol{\omega} \cdot \mathrm{d}\mathbf{S}.$$

The first form is the line integral of velocity along C; the second (Stokes) is the flux of vorticity through any surface spanning C.

(b) Solid-body rotation and line vortex

(i) For solid-body rotation, $\mathbf{u} = \boldsymbol{\Omega} \times \mathbf{r}$. Take a circle of radius r centred on the axis:

$$\Gamma = \oint u_\theta \, r \, \mathrm{d}\theta = \Omega r \cdot 2\pi r = 2\pi \Omega r^2.$$

Apply Stokes with uniform ω:

$$\Gamma = \omega \, \pi r^2.$$

Equate:

$$\omega = 2\Omega.$$

$\boxed{\boldsymbol{\omega} = 2\boldsymbol{\Omega}.}$

(ii) Line vortex $U_\theta = A/r$, $r > 0$. Circulation on a circle of radius r:

$$\Gamma = \oint \frac{A}{r} \, r \, \mathrm{d}\theta = 2\pi A.$$

Vorticity (axisymmetric):

$$\omega_z = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rU_\theta) = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(A) = 0, \qquad r > 0.$$

Comment: vorticity vanishes everywhere except at $r = 0$ but $\Gamma \neq 0$, so all the vorticity is concentrated as a delta function on the axis. The flow is irrotational outside the singular core.

(c) Couette flow between rotating cylinders

Continuity (incompressible, axisymmetric, $U_z = 0$):

$$\frac{1}{r}\frac{\partial(rU_r)}{\partial r} = 0 \;\Rightarrow\; U_r = 0.$$

Symmetry gives $\partial_\theta(\cdot) = 0$ and steady $\partial_t(\cdot) = 0$. Navier–Stokes simplifies; radial component:

$$-\frac{U_\theta^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r}. \tag{1}$$

Tangential component:

$$0 = \nu\left[\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}U_\theta}{\mathrm{d}r}\right) - \frac{U_\theta}{r^2}\right]. \tag{2}$$

Rewrite:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left[\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rU_\theta)\right] = 0.$$

Integrate twice:

$$U_\theta(r) = C_1 r + \frac{C_2}{r}.$$

Boundary conditions (no-slip):

$$U_\theta(r_1) = \Omega_1 r_1, \qquad U_\theta(r_2) = \Omega_2 r_2.$$

Apply BCs:

$$C_1 r_1 + C_2/r_1 = \Omega_1 r_1,$$
$$C_1 r_2 + C_2/r_2 = \Omega_2 r_2.$$

Multiply the first by r_1, the second by r_2:

$$C_1 r_1^2 + C_2 = \Omega_1 r_1^2,$$
$$C_1 r_2^2 + C_2 = \Omega_2 r_2^2.$$

Subtract:

$$C_1(r_2^2 - r_1^2) = \Omega_2 r_2^2 - \Omega_1 r_1^2.$$

Solve for C_1:

$$\boxed{C_1 = \frac{\Omega_2 r_2^2 - \Omega_1 r_1^2}{r_2^2 - r_1^2}.}$$

Substitute back to find C_2:

$$C_2 = \Omega_1 r_1^2 - C_1 r_1^2 = r_1^2\,\frac{(\Omega_1 - \Omega_2) r_2^2}{r_2^2 - r_1^2}.$$

Hence

$$\boxed{C_2 = \frac{(\Omega_1 - \Omega_2)\, r_1^2 r_2^2}{r_2^2 - r_1^2}.}$$

Vorticity (axisymmetric):

$$\omega_z = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rU_\theta) = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(C_1 r^2 + C_2\right).$$

Differentiate:

$$\omega_z = 2C_1.$$

Physical meaning of C_1: the flow consists of a solid-body rotation at angular rate C_1 (carrying uniform vorticity $2C_1$) plus an irrotational line-vortex part C_2/r.

Pressure: integrate (1) with $U_\theta = C_1 r + C_2/r$:

$$U_\theta^2 = C_1^2 r^2 + 2C_1C_2 + \frac{C_2^2}{r^2}.$$

Divide by r:

$$\frac{U_\theta^2}{r} = C_1^2 r + \frac{2C_1C_2}{r} + \frac{C_2^2}{r^3}.$$

Integrate:

$$\boxed{p(r) = p_0 + \rho\left[\tfrac{1}{2}C_1^2 r^2 + 2C_1C_2\ln r - \tfrac{1}{2}\,\frac{C_2^2}{r^2}\right]}\,.$$

(d) Reynolds number and numerics

$Re = \Omega_1 r_1(r_2 - r_1)/\nu$ measures inertial : viscous forces, with the gap $r_2 - r_1$ as the relevant length and $\Omega_1 r_1$ as the relevant velocity. For $Re < 50$ the laminar Couette profile above holds. Increasing Ω_1, $r_2 - r_1$, or decreasing ν raises Re; beyond a critical value the flow undergoes Taylor–Couette instability (Taylor vortices) and eventually turbulence.

Data: $r_1 = 0.25\,\mathrm{m}$, gap $d = r_2 - r_1 = 5\times10^{-3}\,\mathrm{m}$, $\Omega_1 = 20\,\mathrm{rpm} = 20\cdot 2\pi/60 = 2.094\,\mathrm{rad\,s^{-1}}$, $\mu = 0.02\,\mathrm{Pa\,s}$, $\rho = 900\,\mathrm{kg\,m^{-3}}$, hence $\nu = \mu/\rho = 2.22\times10^{-5}\,\mathrm{m^2\,s^{-1}}$.

Reynolds number:

$$Re = \frac{(2.094)(0.25)(5\times10^{-3})}{2.22\times10^{-5}}.$$

Multiply numerator:

$$Re = \frac{2.618\times10^{-3}}{2.22\times10^{-5}}.$$

Evaluate:

$$\boxed{Re \approx 118.}$$

With $C_2 = 0$, $U_\theta = C_1 r$ (pure solid-body rotation between the two cylinders, possible if $\Omega_2 r_2^2/(r_2^2 - r_1^2) = \Omega_1 r_1^2/(\ldots)$, i.e. $\Omega_1 = \Omega_2 \equiv \Omega$). Pressure reduces to

$$p(r) - p(r_1) = \tfrac{1}{2}\rho C_1^2(r^2 - r_1^2) = \tfrac{1}{2}\rho\Omega_1^2(r^2 - r_1^2).$$

Insert $r_2^2 - r_1^2 = (r_2 - r_1)(r_2 + r_1) = (5\times10^{-3})(0.505) = 2.525\times10^{-3}\,\mathrm{m}^2$:

$$\Delta p = \tfrac{1}{2}(900)(2.094)^2(2.525\times10^{-3}).$$

Evaluate the square:

$$\Delta p = \tfrac{1}{2}(900)(4.385)(2.525\times10^{-3}).$$

Multiply:

$$\boxed{\Delta p \approx 4.98\,\mathrm{Pa}.}$$

Interpretation: $Re \approx 118 > 50$, so the laminar approximation is on the borderline of validity — Taylor vortices are likely to appear; the calculated Δp should therefore be regarded only as the laminar reference value.

Question 2 — Bernoulli, Rayleigh–Plesset and bubble collapse

Problem. Derive the vorticity form of momentum and the time-dependent Bernoulli equation, then analyse the collapse of a spherical bubble in an incompressible fluid.

(a) Vorticity form of Navier–Stokes

Navier–Stokes (no gravity):

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\mathbf{u}.$$

Apply the identity $(\mathbf{u}\cdot\nabla)\mathbf{u} = \nabla(\frac{1}{2}u^2) + \boldsymbol{\omega}\times\mathbf{u}$:

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla\big(\tfrac{1}{2}u^2\big) + \boldsymbol{\omega}\times\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\mathbf{u}.$$

Assume ρ constant (incompressible; needed to absorb $1/\rho$ into the gradient):

$$\boxed{\frac{\partial \mathbf{u}}{\partial t} = -\nabla\left(\frac{p}{\rho} + \tfrac{1}{2}u^2\right) + \mathbf{u}\times\boldsymbol{\omega} + \nu\nabla^2\mathbf{u}.}$$

(b) Time-dependent Bernoulli

Irrotational: $\mathbf{u} = \nabla\phi$, $\boldsymbol{\omega} = 0$. Inviscid: $\nu = 0$. Substitute:

$$\nabla\frac{\partial\phi}{\partial t} = -\nabla\left(\frac{p}{\rho} + \tfrac{1}{2}u^2\right).$$

Move all terms under one gradient:

$$\nabla\left(\frac{\partial\phi}{\partial t} + \frac{p}{\rho} + \tfrac{1}{2}u^2\right) = 0.$$

Integrate in space (constant of integration depends only on t):

$$\boxed{\frac{\partial\phi}{\partial t} + \frac{p}{\rho} + \tfrac{1}{2}u^2 = C(t).}$$

(c) Velocity field around the bubble

Spherical symmetry, incompressibility:

$$\frac{1}{r^2}\frac{\partial(r^2u_r)}{\partial r} = 0.$$

Integrate:

$$u_r(r,t) = \frac{f(t)}{r^2}.$$

Boundary condition at the bubble surface $r = R(t)$:

$$u_r(R,t) = \frac{\mathrm{d}R}{\mathrm{d}t}.$$

Determine $f(t)$:

$$f(t) = R^2\,\frac{\mathrm{d}R}{\mathrm{d}t}.$$

Hence

$$u_r(r,t) = \frac{R^2}{r^2}\frac{\mathrm{d}R}{\mathrm{d}t}.$$

Velocity potential from $u_r = \partial_r \phi$:

$$\phi(r,t) = -\int_r^\infty u_r\,\mathrm{d}r' = -\frac{\mathrm{d}R}{\mathrm{d}t}\,R^2\int_r^\infty \frac{\mathrm{d}r'}{r'^2}.$$

Evaluate the integral:

$$\boxed{\phi(r,t) = -\frac{R^2}{r}\frac{\mathrm{d}R}{\mathrm{d}t}.}$$

(The constant of integration is fixed by $\phi(\infty,t) = 0$.)

(d) Equation of motion for $R(t)$

Apply Bernoulli at $r = R$ and $r = \infty$. At infinity: $\phi = 0$, $u = 0$, $p = p_\infty$:

$$C(t) = \frac{p_\infty}{\rho}.$$

At $r = R$: $u = \frac{\mathrm{d}R}{\mathrm{d}t}$, $p = p_0 = 0$. Compute $\partial_t\phi$ from $\phi = -R^2\dot{R}/r$:

$$\left.\frac{\partial\phi}{\partial t}\right|_{r=R} = -\frac{1}{R}\left[2R\dot{R}^2 + R^2\ddot{R}\right] + \frac{R^2\dot{R}}{R^2}\dot{R}\cdot 0.$$

Simplify:

$$\left.\frac{\partial\phi}{\partial t}\right|_{r=R} = -2\dot{R}^2 - R\ddot{R}.$$

Bernoulli at $r = R$:

$$-2\dot{R}^2 - R\ddot{R} + 0 + \tfrac{1}{2}\dot{R}^2 = \frac{p_\infty}{\rho}.$$

Combine:

$$\boxed{R\ddot{R} + \tfrac{3}{2}\dot{R}^2 = -\frac{p_\infty}{\rho}.}$$

This is the Rayleigh–Plesset equation for an empty bubble.

(ii) Multiply by $2R^2\dot{R}$:

$$2R^3\dot{R}\ddot{R} + 3R^2\dot{R}^3 = -\frac{2p_\infty}{\rho}R^2\dot{R}.$$

Recognise the LHS:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\dot{R}^2R^3\right) = 2R^3\dot{R}\ddot{R} + 3R^2\dot{R}^3.$$

Recognise the RHS:

$$-\frac{2p_\infty}{\rho}R^2\dot{R} = -\frac{2p_\infty}{3\rho}\frac{\mathrm{d}}{\mathrm{d}t}(R^3).$$

Equate and integrate from $t = 0$ ($R = R_0$, $\dot{R} = 0$) to t:

$$\dot{R}^2R^3 = -\frac{2p_\infty}{3\rho}(R^3 - R_0^3).$$

Rearrange:

$$\dot{R}^2 = \frac{2p_\infty}{3\rho}\left(\frac{R_0^3}{R^3} - 1\right).$$

Take square root (collapse: $\dot{R} < 0$, but the question asks for the magnitude form):

$$\boxed{\frac{\mathrm{d}R}{\mathrm{d}t} = \sqrt{\frac{2p_\infty}{3\rho}}\sqrt{\frac{R_0^3}{R^3} - 1}, \qquad B = \sqrt{\frac{2p_\infty}{3\rho}}.}$$

(e) Collapse time

Separate variables (taking $\dot{R} < 0$, so $\mathrm{d}t = -\mathrm{d}R/(B\sqrt{R_0^3/R^3 - 1})$):

$$\Delta t = \int_0^{R_0} \frac{\mathrm{d}R}{B\sqrt{R_0^3/R^3 - 1}}.$$

Substitute $x = R/R_0$:

$$\Delta t = \frac{R_0}{B}\int_0^1 \frac{\mathrm{d}x}{\sqrt{x^{-3} - 1}}.$$

Insert B:

$$\boxed{\Delta t = I\sqrt{\frac{3\rho}{2p_\infty}}\,R_0, \qquad I = \int_0^1 \frac{\mathrm{d}x}{\sqrt{x^{-3} - 1}} \approx 0.75.}$$

Numerical: $\rho = 10^3\,\mathrm{kg\,m^{-3}}$, $p_\infty = 10^5\,\mathrm{Pa}$, $R_0 = 10^{-3}\,\mathrm{m}$.

$$\sqrt{\frac{3\rho}{2p_\infty}} = \sqrt{\frac{3 \times 10^3}{2 \times 10^5}} = \sqrt{1.5 \times 10^{-2}}.$$

Evaluate the root:

$$\sqrt{1.5 \times 10^{-2}} = 0.1225\,\mathrm{s\,m^{-1}}.$$

Multiply:

$$\Delta t = (0.75)(0.1225)(10^{-3}).$$

Evaluate:

$$\boxed{\Delta t \approx 9.2 \times 10^{-5}\,\mathrm{s} \approx 92\,\mu\mathrm{s}.}$$

Question 3 — Unforced Duffing oscillator

Problem. A mass on a nonlinear spring with linear drag obeys $m\ddot{x} = -kx - \alpha x^3 - \beta\dot{x}$. Cast as a 2D dynamical system, find fixed points, classify them, sketch phase portraits and the bifurcation diagram, and add weak damping.

(a) Duffing form

Newton:

$$m\ddot{x} = -kx - \alpha x^3 - \beta\dot{x}.$$

Divide by m:

$$\ddot{x} = -\frac{\beta}{m}\dot{x} - \frac{k}{m}x - \frac{\alpha}{m}x^3.$$

Let $y = \dot{x}$:

$$\dot{x} = y,$$
$$\dot{y} = -\frac{\beta}{m}y - \frac{k}{m}x - \frac{\alpha}{m}x^3.$$

Identify

$$\boxed{a = \frac{\beta}{m}, \quad b = \frac{k}{m}, \quad c = \frac{\alpha}{m}.}$$

(b) $a = 0$**,** $b = -c$

Equations:

$$\dot{x} = y, \qquad \dot{y} = -bx - cx^3 = -bx + bx^3 = bx(x^2 - 1).$$

Fixed points $\dot{x} = \dot{y} = 0$: $y = 0$ and $x(x^2 - 1) = 0$:

$$x^* \in \{0, +1, -1\}.$$

Jacobian:

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ -b - 3cx^2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -b + 3bx^2 & 0 \end{pmatrix}.$$

At $(0, 0)$ with $b = -1$ (so $-b = 1$):

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Eigenvalues:

$$\lambda^2 = 1 \;\Rightarrow\; \lambda = \pm 1.$$

Real, opposite sign: **saddle**.

At $(\pm 1, 0)$ with $b = -1$: $-b + 3b(1) = 1 - 3 = -2$:

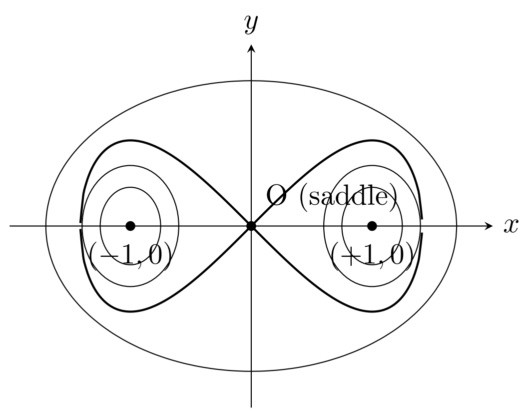
$$J = \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix}.$$

Eigenvalues:

$$\lambda^2 = -2 \;\Rightarrow\; \lambda = \pm i\sqrt{2}.$$

Pure imaginary, $\tau = 0$: **centres** (linear); the conserved Hamiltonian $H = \frac{1}{2}y^2 + \frac{1}{2}bx^2 + \frac{1}{4}cx^4$ confirms they remain centres in the full nonlinear system.

Phase portrait ($b = -1$**,** $c = +1$**, double-well).**



(c) $a = 0$**, general** b, c

Fixed points: $y^* = 0$, $bx + cx^3 = 0$:

$$x^*(b + cx^{*2}) = 0.$$

Always: $x^* = 0$. Additionally, if $-b/c > 0$:

$$x^* = \pm\sqrt{-b/c}.$$

Existence condition: $bc < 0$.

Jacobian eigenvalues: $\lambda^2 = -(b + 3cx^{*2})$.

Origin $(0, 0)$**:** $\lambda^2 = -b$.

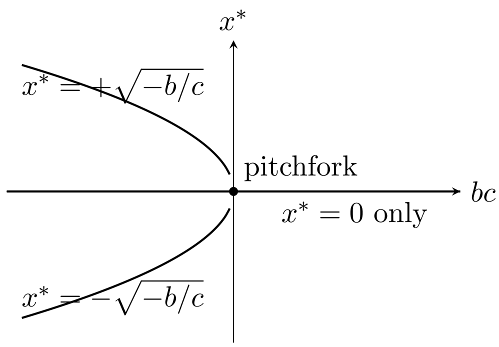
- $b > 0$: $\lambda = \pm i\sqrt{b} \Rightarrow$ centre.
- $b < 0$: $\lambda = \pm\sqrt{-b}$ (real, opposite sign) $\Rightarrow$ saddle.

Side fixed points $x^* = \pm\sqrt{-b/c}$ **(require** $bc < 0$**):** substitute $cx^{*2} = -b$:

$$\lambda^2 = -(b - 3b) = 2b.$$

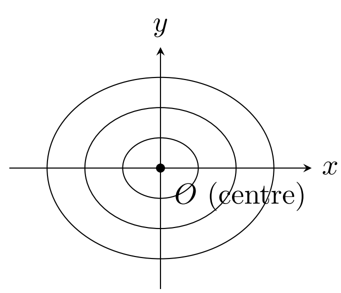
- $b < 0$ $(c > 0)$: $\lambda^2 = 2b < 0$, $\lambda = \pm i\sqrt{-2b} \Rightarrow$ centres.
- $b > 0$ $(c < 0)$: $\lambda^2 = 2b > 0$, $\lambda = \pm\sqrt{2b} \Rightarrow$ saddles.

Bifurcation diagram (x^* **vs** bc**).** A pitchfork bifurcation at $bc = 0$.



Stability legend: solid branches stable centres for $b < 0, c > 0$; the central branch is a saddle in that range and a centre for $b > 0$.

Phase portrait, $b > 0$**.** Take $c > 0$ first (single well). Origin is a centre, no other fixed points; closed orbits encircle the origin.



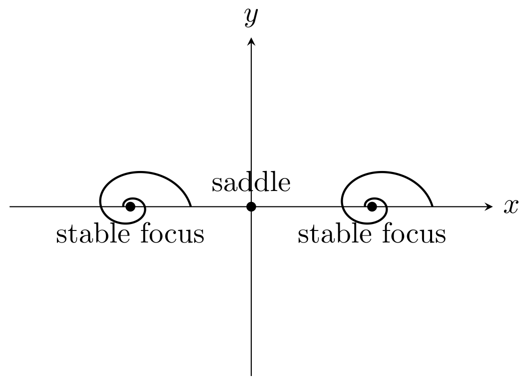
For $b > 0, c < 0$ (still $b > 0$ but inverted quartic): origin is still a centre, and side points $x^* = \pm\sqrt{-b/c} = \pm\sqrt{b/|c|}$ are saddles connected by heteroclinic separatrices that enclose the central centre; orbits inside the separatrix are closed, those outside escape to infinity.

(d) Weak damping, $b < 0$, $a^2 < 4|b|$, fixed points unchanged

Adding $a > 0$ adds $-ay$ to $\dot{y}$; the Jacobian gains $-a$ on the $(2,2)$ entry. Trace $\tau = -a < 0$. At each fixed point, Δ unchanged in sign ($\Delta =$ negative of $J_{21} = b + 3cx^{*2}$ swapped by the off-diagonal structure: $\Delta = b + 3cx^{*2}$).

- Origin (saddle, $b < 0, c > 0$): $\Delta = b < 0$ unchanged $\Rightarrow$ saddle.
- Side wells (centres in (c)): $\Delta = -2b > 0$, $\tau = -a < 0$, discriminant $\tau^2 - 4\Delta = a^2 - 4|2b| < a^2 - 4|b| < 0 \;\Rightarrow$ **stable spirals (foci)**.

Modification of the (b) phase portrait: the two centres at $x = \pm 1$ become inward spirals; the saddle at the origin remains, but its homoclinic separatrices break — trajectories cross between basins only along the unstable manifold of the saddle, otherwise spiral into one well or the other. All orbits eventually reach one of the two stable foci.



Physical interpretation: in the double-well potential $V = \frac{1}{2}bx^2 + \frac{1}{4}cx^4$ ($b < 0$, $c > 0$), the unstable equilibrium at the origin separates two stable equilibria. Without damping, the mass oscillates indefinitely about one minimum or the other (or executes large figure-of-eight orbits along the separatrix). With weak drag, energy is dissipated; the mass spirals into whichever well it ends up in — a bistable mechanical switch.

Question 4 — Freely jointed chain and DNA looping

Problem. Develop the FJC distribution and its entropic-spring limit; obtain the force–extension relation; apply to DNA-looping probability and the optimum loop size.

(a) 3D distribution

Independent Gaussian projections:

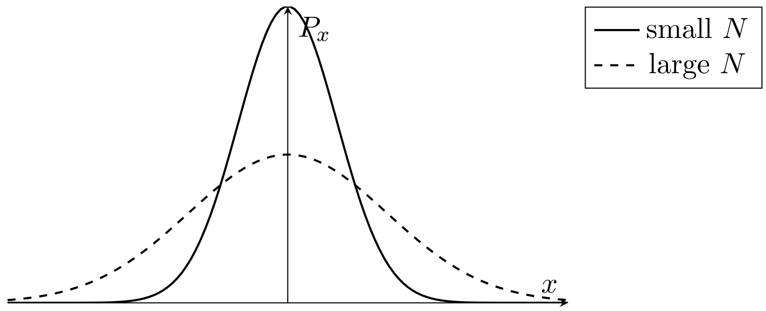
$$P(\mathbf{r}, N) = P_x(x)P_y(y)P_z(z).$$

Multiply:

$$\boxed{P(\mathbf{r}, N) = \left(\frac{3}{2\pi N b^2}\right)^{3/2} \exp\left(-\frac{3r^2}{2Nb^2}\right).}$$

$P_x(x, N)$ is a Gaussian of width $\sigma_x = b\sqrt{N/3}$. As $N \uparrow$, the distribution flattens and broadens with $\sigma_x \propto \sqrt{N}$.

Sketch of $P_x(x, N)$ for two values of N.



Brownian-motion analogy: the chain end position $\mathbf{r}$ after N steps of length b is the sum of independent unit-length random vectors, identical to the position of a random walker after N steps. By the central limit theorem, both produce Gaussian distributions with $\langle r^2 \rangle = Nb^2$; the chain configuration is the spatial counterpart of a Brownian trajectory.

(b) Entropic spring

Boltzmann entropy:

$$S(\mathbf{r}) = k_B \ln P(\mathbf{r}, N) + \text{const.}$$

Substitute the Gaussian:

$$S(\mathbf{r}) = k_B \left[\tfrac{3}{2} \ln\left(\frac{3}{2\pi N b^2} \right) - \frac{3r^2}{2Nb^2} \right].$$

Helmholtz free energy at fixed T (no internal energy, $U = 0$ for FJC):

$$F(\mathbf{r}) = -TS(\mathbf{r}).$$

Substitute:

$$F(\mathbf{r}) = -T\, k_B\, \tfrac{3}{2} \ln\bigl(3/2\pi N b^2\bigr) + \frac{3k_B T}{2Nb^2} r^2.$$

Identify

$$\boxed{F(\mathbf{r}) = d + c\, r^2, \qquad c = \frac{3k_B T}{2Nb^2}, \qquad d = \text{const}(T, N).}$$

F is quadratic in r, so $-\nabla F = -2c\mathbf{r}$ acts as a Hooke restoring force with stiffness $\kappa = 2c = 3k_B T/(Nb^2)$. The chain has no internal energy — its restoring force comes entirely from loss of conformational entropy when extended (an entropic spring).

Temperature dependence: $\kappa \propto T$. The entropic stiffness grows linearly with temperature.

(c) Force–extension under tension f

Single segment $\mathbf{b}_i$ with $|\mathbf{b}_i| = b$ has energy $-fb\cos\theta_i$ when $\mathbf{f} = f\hat{\mathbf{z}}$ acts on the chain end. Single-segment partition function:

$$Z_1 = \int_0^\pi e^{fb\cos\theta/(k_B T)} \sin\theta \,\mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\phi.$$

Substitute $u = \cos\theta$:

$$Z_1 = 2\pi \int_{-1}^{1} e^{au}\,\mathrm{d}u, \qquad a = \frac{fb}{k_B T}.$$

Evaluate:

$$Z_1 = \frac{4\pi \sinh a}{a}.$$

Independent segments:

$$Z = Z_1^N = \left(\frac{4\pi \sinh a}{a}\right)^N .$$

Mean extension along $\mathbf{f}$:

$$\langle z\rangle = k_B T\,\frac{\partial \ln Z}{\partial f} = N\,b\,\frac{\partial \ln(\sinh a/a)}{\partial a}.$$

Differentiate:

$$\frac{\partial}{\partial a}\ln(\sinh a/a) = \coth a - \frac{1}{a}.$$

Hence

$$\boxed{\langle z\rangle = Nb\left(\coth a - \frac{1}{a}\right) \equiv Nb\,\mathcal{L}(a), \qquad a = \frac{fb}{k_B T}.}$$

Low-force limit $a \ll 1$: use $\coth a \approx a/3 + 1/a$:

$$\langle z\rangle \approx Nb\,\frac{a}{3} = \frac{Nb^2 f}{3k_B T}.$$

Solve for f:

$$\boxed{f \approx \frac{3k_B T}{Nb^2}\langle z\rangle.}$$

A linear (Hookean) force–extension with the same stiffness $\kappa = 3k_BT/(Nb^2)$ as in (b).

(d) DNA looping probability

Treat dsDNA as a FJC with statistical segment $b = 2P$ in length units. With $P = 150$ bp and bp step 0.34 nm, $P = 51$ nm and Kuhn length $b = 2P = 102$ nm. For a chain of contour length λ (= total DNA length),

$$N = \frac{\lambda}{b}.$$

Probability that the two ends coincide ($\mathbf{r} = 0$, evaluated as the value of the Gaussian density at the origin):

$$p(\lambda) = P(\mathbf{r} = 0, N) = \left(\frac{3}{2\pi N b^2}\right)^{3/2} .$$

Hence

$$\boxed{p(\lambda) \propto N^{-3/2}.}$$

Entropy difference between the looped state and a free chain:

$$\Delta S = k_B \ln p(\lambda) = k_B\left[\tfrac{3}{2}\ln\left(\frac{3}{2\pi b^2}\right) - \tfrac{3}{2}\ln N\right].$$

Collect constants into C:

$$\boxed{\Delta S = k_B\big(C - \tfrac{3}{2}\ln N\big)\,.}$$

(e) Optimum loop size

Loop free energy = elastic + entropic:

$$\Delta G_{\text{loop}} = \frac{\pi P}{R} k_B T - T\,\Delta S.$$

For a loop of contour length λ closing into a circle of radius R, $2\pi R = \lambda$:

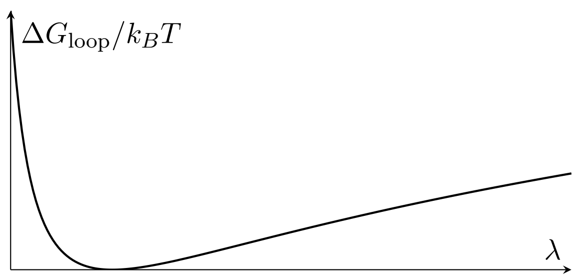
$$R = \frac{\lambda}{2\pi}.$$

Insert R and ΔS:

$$\boxed{\Delta G_{\text{loop}}(\lambda) = \frac{2\pi^2 P}{\lambda} k_B T + \tfrac{3}{2} k_B T \,\ln N - C\, k_B T, \qquad N = \lambda/b.}$$

Behaviour: at small λ, the elastic $1/\lambda$ term dominates — bending is prohibitive. At large λ, the $\frac{3}{2}\ln N$ term dominates — entropic cost of bringing distant ends together. There is a minimum at intermediate length.

Sketch.



Minimise by $\mathrm{d}\Delta G_{\text{loop}}/\mathrm{d}\lambda = 0$:

$$-\frac{2\pi^2 P}{\lambda^2} k_B T + \frac{3 k_B T}{2\lambda} = 0.$$

Solve:

$$\lambda_{\text{min}} = \frac{4\pi^2 P}{3}.$$

Numerical with $P = 51\,\text{nm}$:

$$\lambda_{\text{min}} = \frac{4\pi^2}{3} \times 51\,\text{nm}.$$

Evaluate the prefactor:

$$\frac{4\pi^2}{3} \approx 13.16.$$

Multiply:

$$\boxed{\lambda_{\text{min}} \approx 670\,\text{nm} \approx 2000\,\text{bp}.}$$